



Choosing Your Capstone Domain

Your capstone project is your opportunity to apply comprehensive AI infrastructure knowledge across three potential domains: **NLP**, **Computer Vision**, or **Generative AI**. The right choice balances your interests, project feasibility, and infrastructure skills.

Why Domain Choice Matters



Infrastructure Requirements

Your domain defines specific data needs, infrastructure scale, and tooling requirements that will shape your entire implementation approach.



Performance Metrics

Each domain prioritizes different metrics - from accuracy and latency to cost efficiency and compliance considerations.



Career Portfolio

Your choice creates a showcase piece that signals your expertise and desired career direction to potential employers.

The complexity of your implementation will vary dramatically based on deployment needs (edge vs. cloud) and processing requirements (real-time vs. batch).

Option 1: NLP Infrastructure

Common Use Cases

- Intelligent chatbots and conversational agents
- Document search and semantic retrieval
- Automated text summarization
- Regulatory compliance monitoring

Infrastructure Requirements

- Robust tokenization pipelines
- Vector databases (FAISS, Pinecone, Weaviate)
- GPU resources for transformer models
- MLOps systems for continuous retraining



Key Metrics: Response latency and retrieval accuracy

Option 2: Computer Vision Infrastructure



Computer vision requires significant compute resources and often edge deployment

Common Use Cases

- Medical imaging analysis
- Manufacturing defect detection
- Autonomous navigation systems

Infrastructure Requirements

- Large GPU/TPU clusters for CNNs and Vision Transformers
- Sophisticated image preprocessing pipelines (ETL)
- High-capacity storage for terabytes of images/videos
- Edge deployment solutions for cameras and IoT devices

Key Metric: Inference speed and accuracy on edge devices



Option 3: Generative AI Infrastructure

Use Cases

- Large language models (LLMs)
- Diffusion models for image generation
- Creative content production
- Enterprise copilots and assistants

Infrastructure Needs

- High-performance compute clusters
- Efficient serving infrastructure (sharding, caching)
- Cost optimization techniques for scale
- Safety guardrails for generated outputs

Key Metrics: Generation quality and cost per inference

Comparing Domains

Domain	Data Type	Infra Scale	Deployment	Key Challenges
NLP	Text	Medium-High	Cloud-heavy	Latency, context length
Vision	Images/Video	High	Cloud + Edge	Storage, real-time inference
GenAI	Multi-modal	Very High	Cloud-dominant	Cost, safety, scalability

Each domain represents a different set of infrastructure challenges and optimization priorities. Your choice will determine which technical skills you'll develop most deeply during your capstone project.

Factors in Choosing Your Domain



The ideal capstone domain sits at the intersection of your passion, practical constraints, career aspirations, and what you can reasonably accomplish within your timeframe while still creating something impressive.

Case Study Examples



NLP: Legal Document System

A retrieval-augmented generation (RAG) infrastructure for summarizing and analyzing legal documents with robust vector search capabilities and compliance guardrails.



Vision: Healthcare Imaging

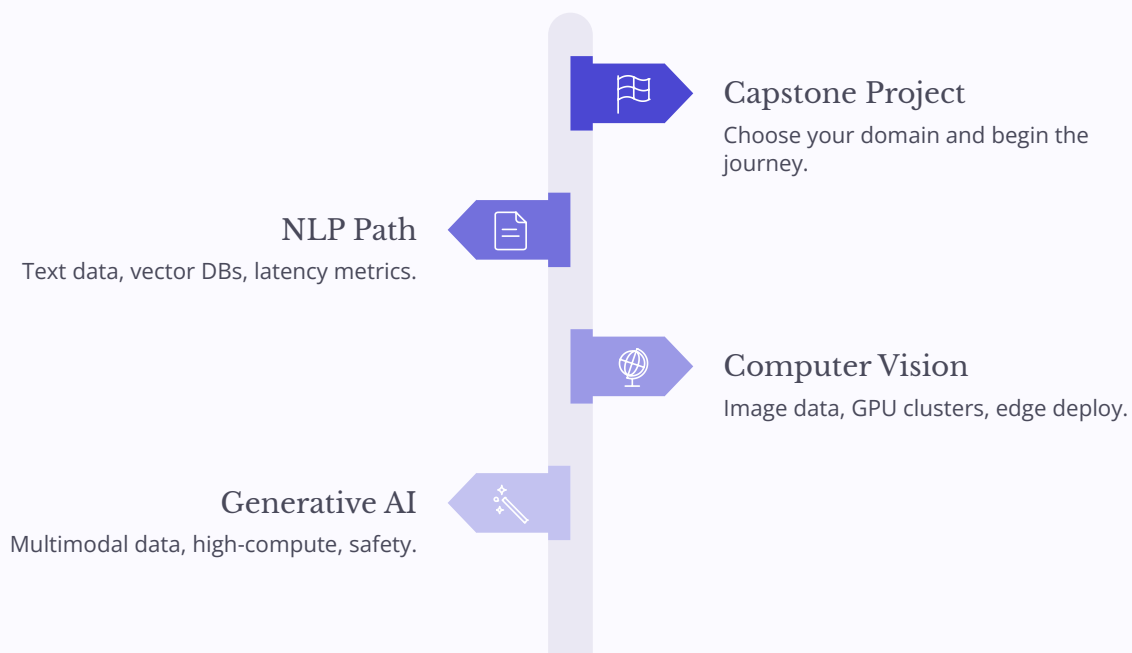
A hybrid cloud pipeline for processing medical images with on-premise edge computing for initial analysis and cloud infrastructure for deeper processing and storage.



GenAI: Enterprise Assistant

A fine-tuned LLaMA model deployed as an enterprise chatbot with cost-optimized inference infrastructure and comprehensive safety filtering.

Domain Choice Sets Your Journey



Your domain choice fundamentally shapes your technical focus, resource requirements, and evaluation criteria.

Key Takeaways

NLP Infrastructure

Focus on data pipelines, vector databases, and cloud APIs. Optimize for response speed and retrieval quality in text-processing systems.

Vision Infrastructure

Build GPU-heavy, storage-intensive systems with edge integration capabilities. Balance resource usage with real-time processing needs.

GenAI Infrastructure

Design compute-hungry, cost-sensitive systems with robust safety mechanisms. Focus on efficient serving at scale for resource-intensive models.

Make Your Choice

Select a domain that aligns with your technical skills, career aspirations, and project feasibility constraints.

Discussion: What's Your Domain?

If you were building your Capstone, would you choose **NLP** (structured text), **Vision** (visual data), or **GenAI** (multi-modal creativity) — and why?

Consider which domain:

- Excites you most technically
- Offers the most career potential
- Provides appropriate challenge level

